

MAT 426, Class 19 Homework: Quadratic Forms & Singular Values

(Due Tuesday, April 21, 2026)

- Find an orthogonal matrix P such that the change of variables $\mathbf{x} = P\mathbf{y}$ transforms the given quadratic form into one with no cross-product terms. Give P and the new quadratic form.
 - $2x_1^2 + 5x_2^2 - 4x_1x_2$
 - $7x_1^2 + x_2^2 + x_3^2 + 8x_1x_2 + 8x_1x_3 - 16x_2x_3$
 - $x^2 + z^2 - 2xy + 2yz$
- Let A be an $n \times n$ symmetric matrix. Prove that the quadratic form $Q(\mathbf{x}) = \mathbf{x}^T A \mathbf{x}$ is
 - positive definite if and only if all of the eigenvalues of A are positive.
 - positive semidefinite if and only if all of the eigenvalues of A are nonnegative.
 - negative definite if and only if all of the eigenvalues of A are negative.
 - negative semidefinite if and only if all of the eigenvalues of A are nonpositive
 - indefinite if and only if A has both positive and negative eigenvalues.
- Let $Q(\mathbf{x}) = \mathbf{x}^T A \mathbf{x}$ be a quadratic form with associated $n \times n$ symmetric matrix A . Let the eigenvalues of A be $\lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_n$. Prove the following subject to the constraint $\|\mathbf{x}\| = 1$
 - $Q(\mathbf{x}) \geq \lambda_n$
 - The maximum value of $Q(\mathbf{x})$ is λ_1 and it occurs when $\mathbf{x}$ is a unit eigenvector corresponding to λ_1 .
 - The minimum value of $Q(\mathbf{x})$ is λ_n and it occurs when $\mathbf{x}$ is a unit eigenvector corresponding to λ_n .
- Consider the conic section $5x^2 + 4xy + 2y^2 = 6$.
 - Sketch the ellipse $x^2 + 6y^2 = 6$ or equivalently $\frac{x^2}{6} + y^2 = 1$.
 - Find the diagonalization of the matrix A associated with the quadratic form on the left side of the equation of the conic section. Ensure that the determinant of P is one; this means, that P corresponds to a rotation because every 2×2 orthogonal matrix with determinant 1 corresponds to a rotation in the xy -plane.
 - Use this diagonalization to perform a change of variables that yields a quadratic form with no cross-product terms.
 - If P corresponds to a rotation and $Q(\mathbf{x}) = \mathbf{x}^T A \mathbf{x} = \mathbf{y}^T D \mathbf{y}$, how does the conic section $5x^2 + 4xy + 2y^2 = 6$ correspond to the conic section $\frac{x^2}{6} + y^2 = 1$?
 - Compute $P\mathbf{e}_1$ and $P\mathbf{e}_2$. Use this, along with the previous observations and sketches, to sketch the conic section $5x^2 + 4xy + 2y^2 = 6$. Note this exercise illustrates why the Principal Axes Theorem is so named.

5. Classify the following quadratic forms as positive definite, positive semidefinite, negative definite, negative semidefinite, or indefinite.

(a) $x_1^2 + x_2^2 - 2x_1x_2$

(b) $-2x^2 - 2y^2 + 2xy$

(c) $2x_1^2 + 2x_2^2 + 2x_3^2 + 2x_1x_1 + 2x_1x_3 + 2x_2x_3$

(d) $x^2 + y^2 - z^2 + 4xy$

6. Let B be an invertible matrix. Show that B^TB is positive definite.

7. Consider the quadratic form $Q(\mathbf{x}) = \mathbf{x}^T A \mathbf{x}$ when $A = \begin{bmatrix} a & b \\ b & d \end{bmatrix}$ and $\det(A) \neq 0$.

(a) If λ_1 and λ_2 are eigenvalues of A , then the characteristic polynomial of A can be written in two ways: $\det(A - \lambda I)$ and $(\lambda - \lambda_1)(\lambda - \lambda_2)$. Use this fact to show that $\lambda_1 + \lambda_2 = a + d$ and $\lambda_1\lambda_2 = \det(A)$.

(b) Prove the following:

i. If $\det(A) > 0$ and $a > 0$, then Q is positive definite.

ii. If $\det(A) > 0$ and $a < 0$, then Q is negative definite.

iii. If $\det(A) < 0$, then Q is indefinite.

8. Show that if an $n \times n$ matrix A is positive definite, then there exists a positive definite matrix B such that $A = B^TB$. (*Hint:* Write $A = PDP^T$ with $P^T = P^{-1}$. Produce a diagonal matrix C such that $D = C^TC$ and let $B = PCP^T$. Show that this B works.)

9. Find the singular values of the matrix A below.

(a) $A = \begin{bmatrix} 1 & 0 \\ 0 & -3 \end{bmatrix}$

(b) $A = \begin{bmatrix} 3 & 0 \\ 8 & 3 \end{bmatrix}$

(c) $A = \begin{bmatrix} -3 & 1 \\ 6 & -2 \\ 6 & -2 \end{bmatrix}$